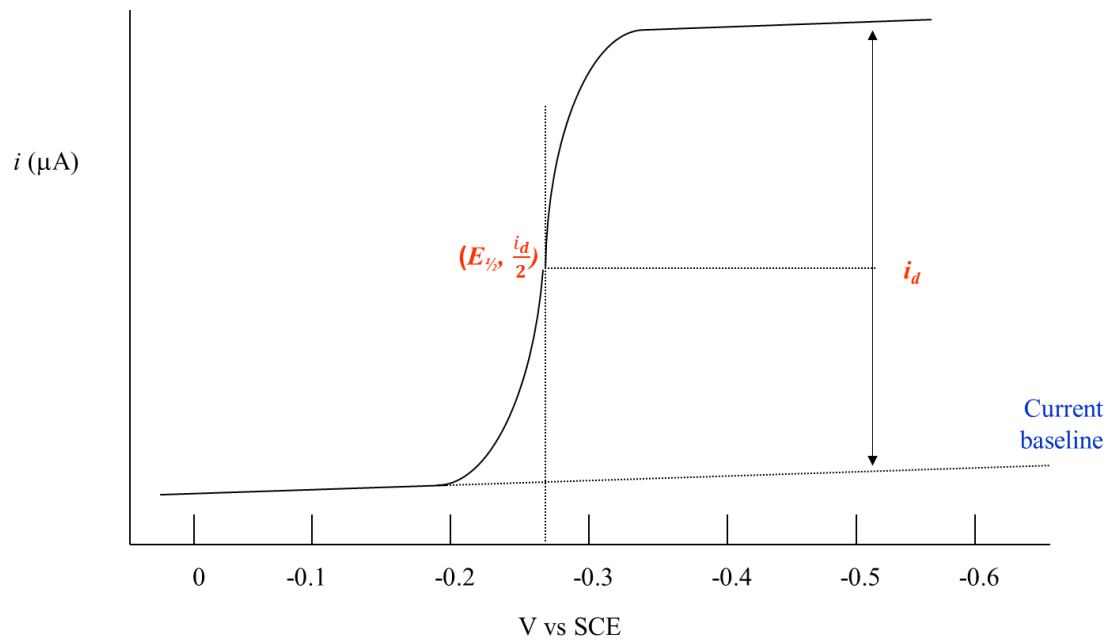
125101 - Analytical chemistry 1 for engineers

Tutorial 11 – Electroanalysis

T11-1. The drug Librium gives a polarographic wave with $E_{1/2} = -0.265\ V$ (versus S.C.E.) in 0.05 M H_2SO_4. A 50.0mL sample containing Librium gave a wave height of 0.37μA. When 2.00 mL of 3.00 mM Librium in 0.05 M H_2SO_4 were added to the sample, the wave height increased to 0.8 μA. Find the molarity of Librium in the unknown.



For quantitative analysis by voltammetry, the diffusion current is proportional to the bulk concentration of analyte if there is a sufficient concentration of supporting electrolyte. The half-wave potential ($E_{1/2}$, the potential at which half the maximum current) is characteristic of a particular analyte in a particular medium. $E_{1/2}$ can be used for qualitative analysis.

We set the molarity of Librium in the unknown is x,

$$\text{So, } y_1 = 0.37\ \mu A = k\,x$$

The concentration after the addition is

$$\frac{50.0\ mL\ x\ + 2.00\ mL \times 3.00\ mM}{(50.0 + 2.00)\ mL}$$

$$y_2 = k\frac{50.0\ mL\ x + 2.00\ mL \times 3.00\ mM}{(50.0+2.00)\ mL} = 0.8\ \mu A$$

Then, we can obtain x according to these two equations.

$$x = 0.096\ mM$$

Actually, the voltammetry and standard addition are combined in this question.

T11-2. Ions that react with Ag^+ can be determined electrogravimetrically by deposition on a silver working anode:

$$Ag(s) + X^- \longrightarrow AgX(s) + e \qquad X = Cl^-, Br^-, I^-, SCN^-$$

a. What will be the final mass of a silver anode used to electrolyze 75.00 mL of 0.02380 M $KSCN$ (M.W. of SCN is 58.078 g/mol) if the initial mass of the Ag anode is 12.4683g?

b. At what electrolysis voltage (versus S.C.E. with E = 0.241 V) for $AgBr(s)$ will be deposited from 0.10 M Br^-? Consider negligible current flow, so that there is no ohmic potential, concentration polarization, or over potential.

$$AgBr(s) + e \rightleftharpoons Ag(s) + Br^- \text{ with } E^o = 0.071V$$

c. Is it theoretically possible to separate 99.99% of 0.1 M KI from 0.10 M KBr by controlled potential electrolysis?

$$Ksp(AgI) = 8.3 \times 10^{-17}$$

$$Ksp(AgBr) = 5.0 \times 10^{-13}$$

a) 75.00 mL of 0.02380 M KSCN = 1.785 mmol of SCN^-, which gives 1.785 mmol of AgSCN

$$Ag(s) + SCN^- \longrightarrow AgSCN(s) + e$$

The initial mass of the Ag anode is 12.4683 g, the mass of SCN is 1.785 mmol × 58.078 g/mol = 103.67 mg.

Therefore, the final mass of a silver anode is 12.4683 g + 0.1037 g = 12.5675 g.

b) Anode: $Ag(s) + Br^- \longrightarrow AgBr(s) + e$

Write it in reduction way for Nernst equation: $AgBr(s) + e \rightleftharpoons Ag(s) + Br^-$ $E^o = 0.071$ V

E(Anode) = 0.071 – 0.05916 log [Br^-]

= 0.071 – 0.05916 log0.10 = 0.130 V

E(Cathode) = E(S.C.E.) = 0.241 V

E = E(Cathode) - E(Anode) = 0.241 – 0.130 = 0.111 V

So, electrolysis voltage of 0.111 V (versus S.C.E.) will make AgBr(s) be deposited on Ag electrode from 0.10 M Br^-.

c) Removal of 99.99% of 0.1 M KI will leave

$[I^-] = 1.0 \times 10^{-5}$ M

The $[Ag^+]$ in equilibrium with such I^- is

$[Ag^+] = K_{sp}(AgI)/ [I^-] = 8.3 \times 10^{-12}$ M

The $[Ag^+]$ in equilibrium with 0.10 M KBr is

$[Ag^+]' = K_{sp}(AgBr)/ [Br^-] = 5.0 \times 10^{-12}$ M

Therefore, 8.3×10^{-12} M of Ag^+ will begin to deposit 0.10 M Br-. In theory, the separation is **<u>not possible</u>** by controlled potential electrolysis.

T11-3. A constant current of 0.800 A is used to deposit copper (M.W. of Cu is 63.55 g/mol) at the cathode and release oxygen at the anode of an electrolytic cell. Calculate the mass (in grams) of each product formed in 15.2 min. Assume that no other redox reaction occurs:

$$Cu_{(aq)}^{2+} + 2e \rightleftarrows Cu_{(s)}$$

$$2H_2O_{(l)} \rightleftarrows O_{2(g)} + 4H^+_{(aq)} + 4e$$

M.W. of Cu is 63.55 g/mol; M.W. of O_2 is 32 g/mol.

$Q = n_e \times F = I \times t$

Q – electric charge (Coulomb, C);

n_e – mole of electrons (mol)

F – Faraday constant, 9.6485×10^4 C/mol

I – current (Ampere, A=$\frac{C}{s}$)

t – time (Second, s)

So, $n_e = \frac{I \times t}{F} = \frac{0.8 \times 15.2\times60}{9.6485\times10^4}$ mol = 7.56×10^{-3} mol *e*

At cathode, reduction reaction occurs, the product is Cu. For each Cu^{2+} reacts with 2*e*, so the mole of Cu is half of mole of electrons.

The mass of Cu is:

$$(\frac{1}{2}\times7.56\times10^{-3}) \text{ mol} \times 63.55 \text{ g/mol} = 0.240 \text{ g Cu}$$

At anode, oxidation reaction occurs, the product is O_2. For each O_2 reacts with $4e$, so the mole of O_2 is ¼ of mole of electrons.

The mass of O_2 is:

$$(\frac{1}{4}\times 7.56\times 10^{-3})\ \text{mol} \times 32\ \text{g/mol} = 0.0605\ \text{g}\ O_2$$

T11-4. The same set-up in **T11-3** was used for a **coulometric titration**. Several drops of Bromothymol blue indicator (pK_a = 7.1, pH range of 6.0 to 7.6 with the color change from yellow to blue) and 1.00 mL without knowing concentration KOH in the anode cell. Press the start button, to generate acid in the anode cell by circulating constant current of 0.800 A. When you observe color in the solution changing from yellow to blue, turn off the current and count the time of 15.2 min for the reactions. Try to compute the concentration of KOH.

$Q = n_e \times F = I \times t$

So, $n_e = \frac{I \times t}{F} = \frac{0.8 \times 15.2\times 60}{9.6485\times 10^4}$ mol = 7.56×10^{-3} mol *electrons*

At anode, oxidation reaction occurs, the product is H^+. which will be employed to titrate the KOH.

According to the equation, $2H_2O(l) \rightleftarrows O_2(g) + 4H^+(aq) + 4e$, 1 mole electron will also obtain 1 mole of H^+, so the mole of H^+ equals to the mole of electrons 7.56×10^{-3} mol.

$$OH^- + H^+ \rightleftarrows H_2O(l)$$

The mole of KOH equals to the mole of H^+ in the coulometric titration of acid-base.

The concentration of KOH is:

$$7.56\times 10^{-3}\ \text{mol} \div 1.00\ \text{mL} = 7.56\ \text{mol/L KOH}$$